

Non-Stationary and Seasonality Models

Chapter 4

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Outline

- The Unit-Root Case
- Testing for Unit Roots
- ARIMA Models
- ARIMA Forecasting
- SARIMA Models
- The ETS-ARIMA Duality

The Unit-Root Case

Section 1

The Unit-Root Case

- ARMA(p,q) models assume series is stationary
 - All characteristic roots fall inside the unit-circle
- What if $\{y_t\}$ is a unit-root process?

ACF	Does not decay – stays near 1 for many lags
Yule-Walker	R becomes singular: Can't estimate the model
MLE likelihood	Standard asymptotic theory no longer applies
Forecasts	No mean-reversion, series starts to drift

The Unit-Root Case

- The **autoregressive** AR(1) process
 - $y_t = \phi y_{t-1} + \bar{y} + \varepsilon_t; \quad \varepsilon_t \sim WN(0,1)$
 - Take y_t and subtract its own memory; what is left is a constant + a random shock
 - $\underbrace{y_t - \phi y_{t-1}}_{\text{explained}} = \underbrace{c + \varepsilon_t}_{\text{stochastic}}$
 - Remember the stability condition:
 - $|\phi| < 1 \rightarrow$ stable
 - $|\phi| = 1 \rightarrow$ unit-root process
 - $|\phi| > 1 \rightarrow$ unstable (explosive behavior)

The Unit-Root Case

- The simple AR(1) case
 - $y_t = y_{t-1} + \varepsilon_t$
- Rearrange and apply the lag operator
 - $y_t - y_{t-1} = \varepsilon_t$
 - $(1 - L)y_t = \varepsilon_t$
 - $\Delta y_t = \varepsilon_t$
- Note the following:
 - y_t is non-stationary
 - Δy_t is stationary

The Unit-Root Case

- Unit root without drift ($\bar{y} = 0$)
 - $\Delta y_t = \varepsilon_t$
 - The **level** forecast is **flat** at every horizon
 - $y_{t+1} = y_t + \varepsilon_{t+1}$
 - $y_{t+1} = y_{t-1} + (\varepsilon_{t+1} + \varepsilon_t)$
 - $y_{t+1} = y_0 + (\sum_{t=0}^{t+1} \varepsilon_t)$

The Unit-Root Case

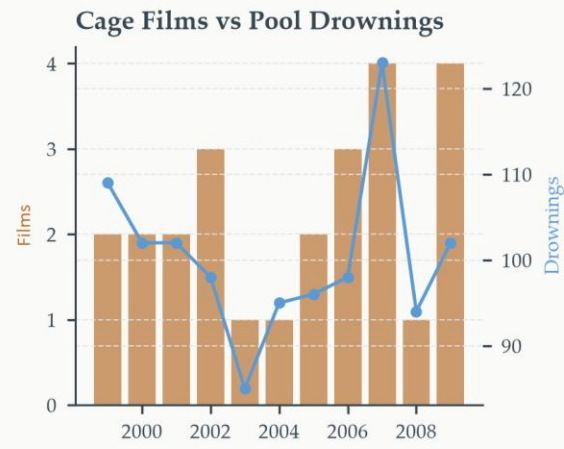
- Unit root with drift ($\bar{y} \neq 0$)
 - $\Delta y_t = \bar{y} + \varepsilon_t$
 - The **level** forecast **grows linearly** at rate $\bar{y}$
 - $y_{T+1} = y_T + \bar{y} + \varepsilon_{t+1}$
 - $y_{T+1} = y_{T-1} + \bar{y} + \bar{y} + (\varepsilon_{t+1} + \varepsilon_t)$
 - $y_{t+1} = y_{T-1} + 2\bar{y} + (\varepsilon_{t+1} + \varepsilon_t)$
 - $y_{T+1} = y_0 + (T + 1)\bar{y} + (\sum_{t=0}^{T+1} \varepsilon_t)$
 - $y_{T+n} = y_0 + (T + n)\bar{y} + (\sum_{t=0}^{T+n} \varepsilon_t)$

The Unit-Root Case

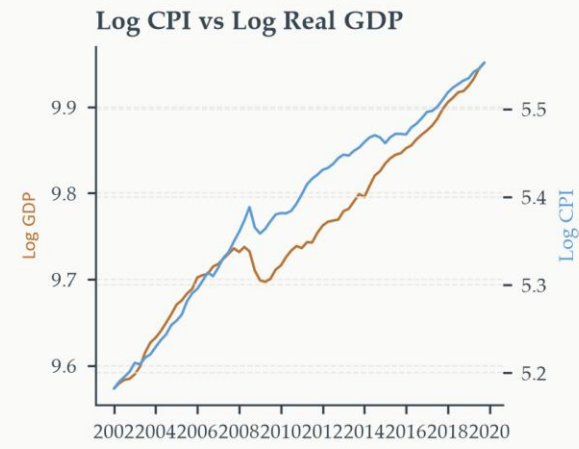
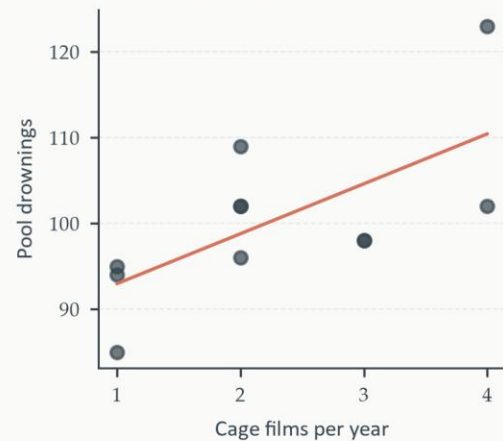
- Spurious regression
 - Assume two unrelated random walk processes with a similar drift
 - Then regressions find “artificial” relationships
 - If two independent series drift, then they will appear correlated in any finite sample just by construction
 - **Obvious** spurious regression
 - Nicolas Cage film appearances correlate with swimming pool drownings ($r = 0,67, t > 2, p(t) < 0.05, R^2 > 0$)
 - What if the spurious regression is **not obvious**?
 - Regress log of CPI on log of real GDP
 - Is this spurious? Why?

The Unit-Root Case

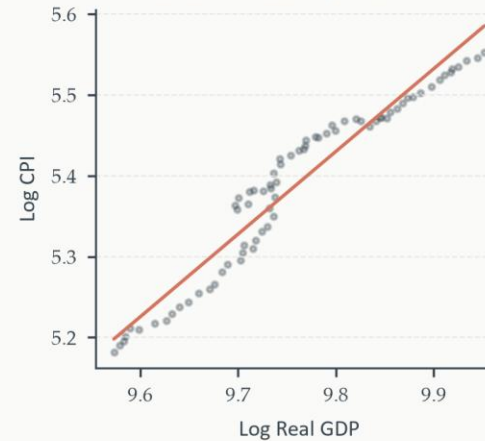
Spurious Regressions: Obvious (left) and Plausible (right)



$$R^2 = 0.44, t = 3.0, p = 0.025$$



$$R^2 = 0.93, t = 31.4, p = 0.0000$$



Testing for Unit Roots

Section 2

Testing for Unit Roots

- Always start with visual inspection
 - You can visually confirm non-stationarity
 - You cannot visually confirm stationarity

Testing for Unit Roots

- The ADF Test (review from chapter 1)

$$\Delta y_t = \delta y_{t-1} + \sum_{j=1}^p \psi_j \Delta y_{t-j} + \alpha + \beta t + \mu_t$$

- $H_0: \delta = (\phi - 1) = 0$ (unit root)
- $H_1: \delta = (\phi - 1) < 0$ (stationarity)
- Test distribution: Dickey-Fuller distribution

Testing for Unit Roots

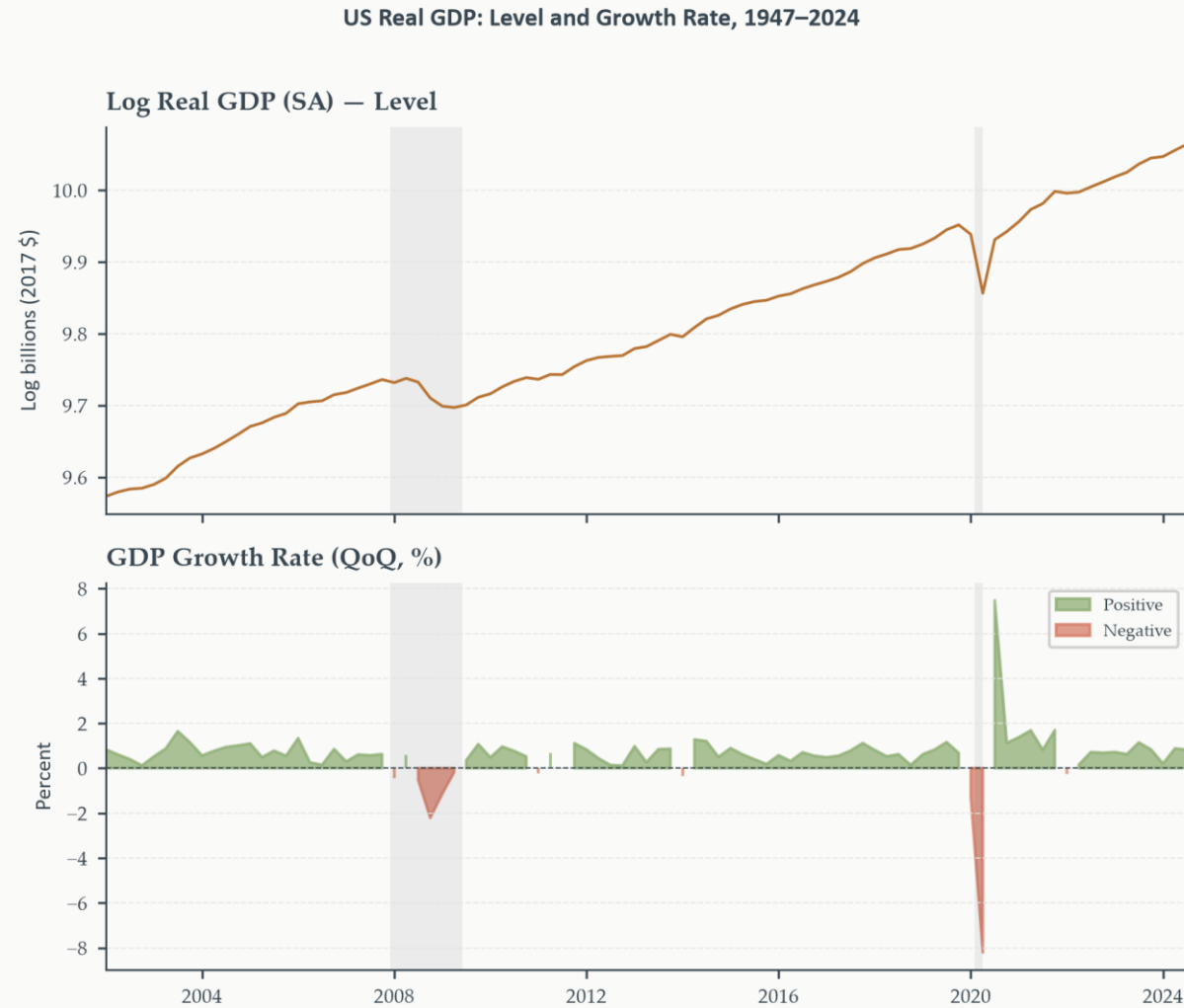
- The KPSS Test (review from chapter 1)

$$y_t = \xi t + r_t + \varepsilon_t \text{ where } r_t = r_{t-1} + \mu_t \text{ and } u_t \sim WN$$

$$KPSS = \frac{1}{T^2} \frac{\sigma_{t=1}^T S_t^2}{\hat{\lambda}(l)^2}; \quad \left(S_t^2 = \sum_{s=1}^t \hat{\varepsilon}_s \right)$$

- $H_0: \sigma_u^2 = 0$ (stationary) $\rightarrow$ then r_t is deterministic and u_t is “inside” ε_t
- $H_1: \sigma_\mu^2 > 0$ (non-stationary)
- Test distribution: KPSS critical values

Testing for Unit Roots



Log SA Real GDP – Level

ADF (H_0 : unit root | spec: 'ct' | p_max=11)

Selection	Lags	Stat	p-val	
AIC	0	-2.6256	0.2683	← fail to reject H_0
BIC	0	-2.6256	0.2683	← fail to reject H_0

Critical values:

1% : -4.0631
5% : -3.4605
10% : -3.1563

KPSS (H_0 : stationary | spec: 'ct')

Bandwidth	nlags	Stat	p-val	
Auto	5	0.2021	0.0152	← reject H_0
Reduced (½)	2	0.3479	0.0100	← reject H_0
Expanded (×2)	10	0.1378	0.0653	

Critical values (auto bandwidth):

10% : 0.1190
5% : 0.1460
2.5% : 0.1760
1% : 0.2160

GDP Growth Rate (QoQ)

ADF (H₀: unit root | spec: 'c' | p_max=11)

Selection	Lags	Stat	p-val
AIC	0	-11.3400	0.0000 ← reject H ₀
BIC	0	-11.3400	0.0000 ← reject H ₀

Critical values:

1% : -3.5052
5% : -2.8942
10% : -2.5842

KPSS (H₀: stationary | spec: 'c')

Bandwidth	nlags	Stat	p-val
Auto	5	0.0894	0.1000
Reduced (½)	2	0.0724	0.1000
Expanded (×2)	10	0.1056	0.1000

Critical values (auto bandwidth):

10% : 0.3470
5% : 0.4630
2.5% : 0.5740
1% : 0.7390

Testing for Unit Roots

- Integration order
 - $y_t \sim I(d)$ if $\Delta^d y_t$ is covariance-stationary but $\Delta^{d-1} y_t$ is not
 - $I(0)$: stationary in levels
 - $I(1)$: first-difference stationary (the majority of economic series)
- $y_t = y_{t-1} + \bar{y} + \varepsilon_t \rightarrow$ non-stationary
- $\Delta y_t = \bar{y} + \varepsilon_t \rightarrow$ stationary

ARIMA Models

Section 3

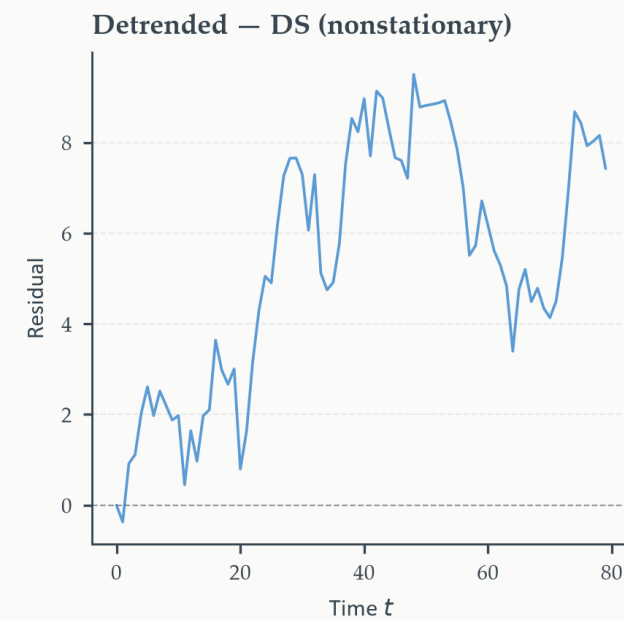
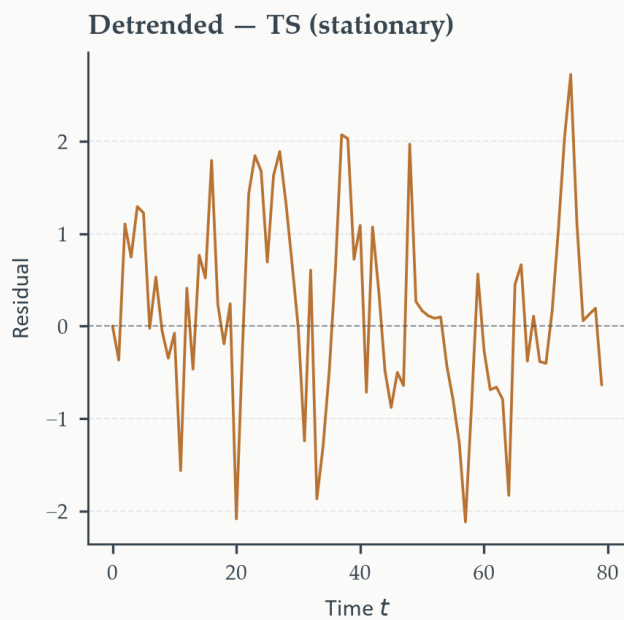
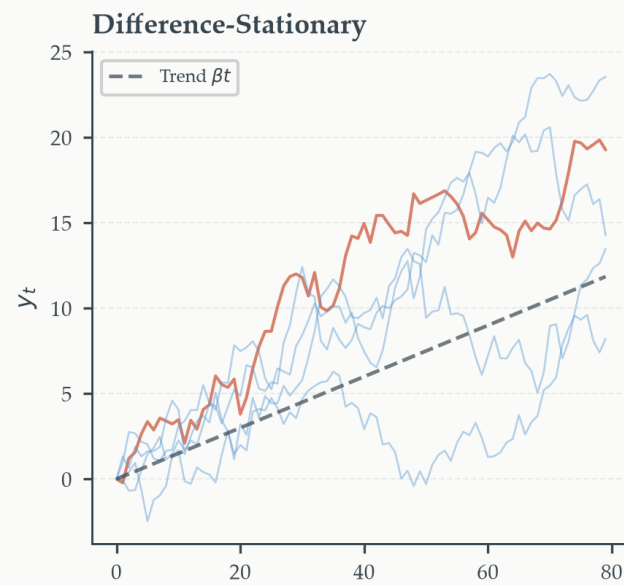
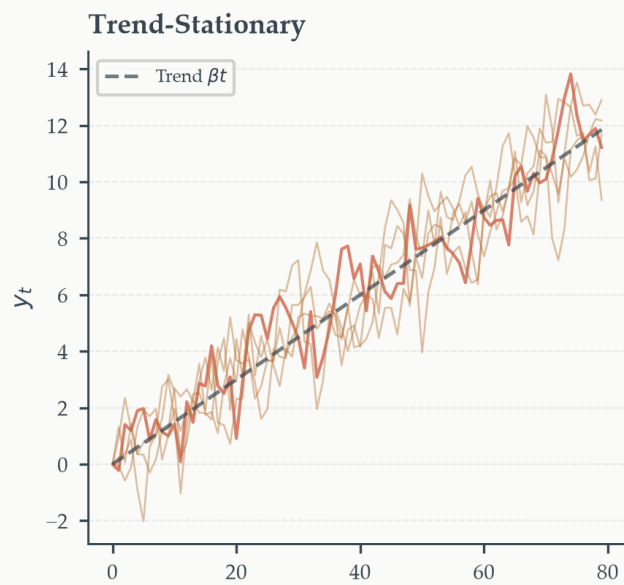
ARIMA Models

- The ARIMA(p,d,q) model is an ARMA model on a series integrated d times
- Example: ARIMA(p,1,q)
 - $$\Delta y_t = \underbrace{(\phi_1 \Delta y_{t-1} + \cdots + \phi_p \Delta y_{t-p})}_{AR(p)} + \bar{y} + \underbrace{(\varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q})}_{MA(q)}$$

ARIMA Models

- Differencing vs Detrending
 - Trend-stationary
 - $y_t = \alpha + \beta t + \varepsilon_t$
 - In this case, you have an ARMA(p,q) with trend
 - Difference-stationary
 - $y_t = y_{t-1} + \bar{y} + \varepsilon_t$
 - No trend to get back to; each shock permanently shifts the level
 - Adding the βt (deterministic) term does not fix this problem (in statistical terms, you have a stochastic trend)
- How do you decide what do you have?
 - What does theory inform you..?
 - ADF test specifications...

Trend-Stationary vs Difference-Stationary: Same Drift, Same Noise



ARIMA Models

- The Box-Jenkins Workflow
 - **Step 0:** Check for stationarity (now you can do this!)
 - **Step 1:** Initial identification. Inspect ACF/PACF and select the best visual candidate for p and q
 - **Step 2:** Fit the model (estimate ϕ and θ) by maximum likelihood
 - **Step 3:** Model diagnostics. Check if residuals behave as WN. If they don't, return to Step 1
 - Collect candidate models by iterating Steps 1 through 3
 - **Step 4:** Model selection. Choose your best model, typically by information criteria and parsimony
 - **Step 5:** Forecasting! 😊

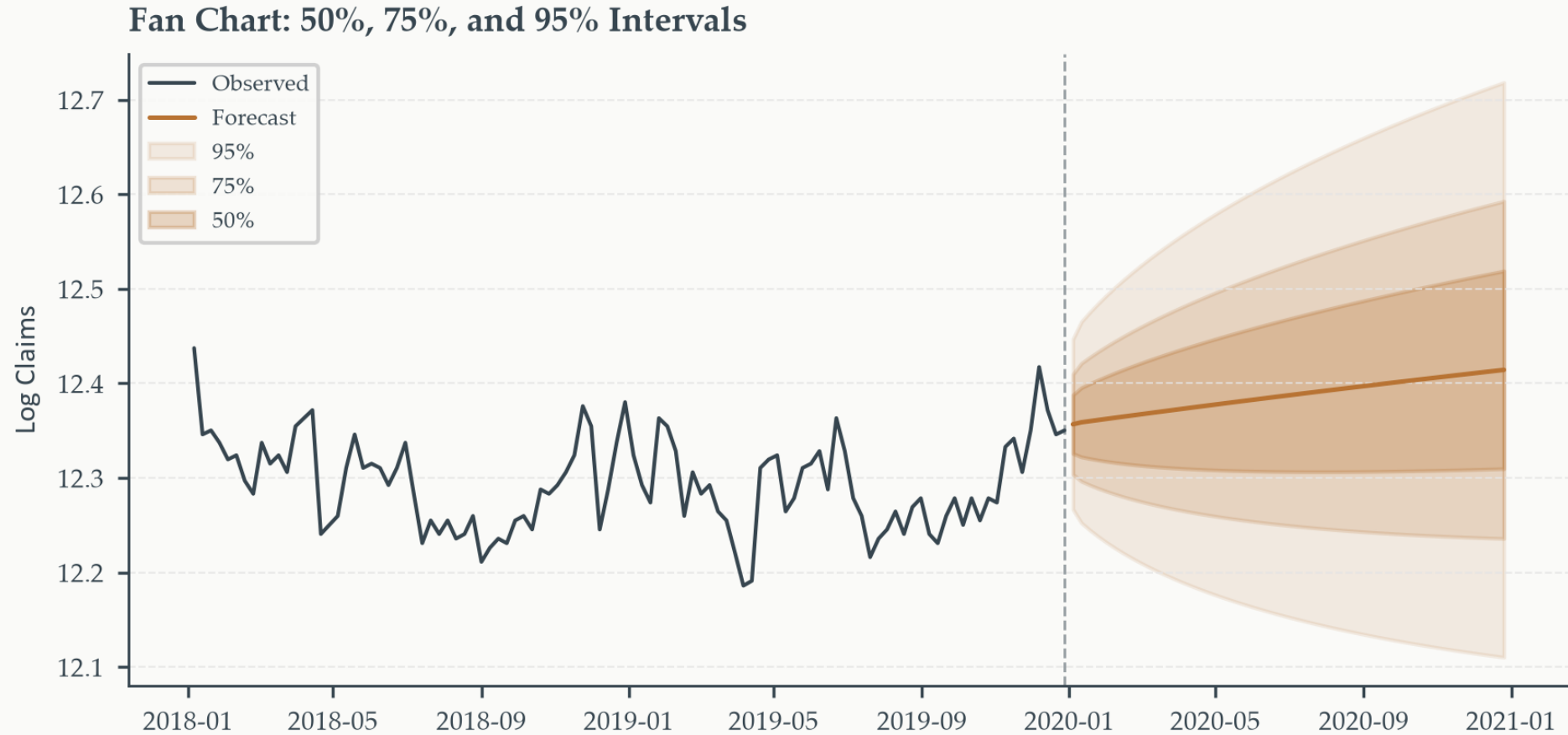
ARIMA Forecasting

Section 4

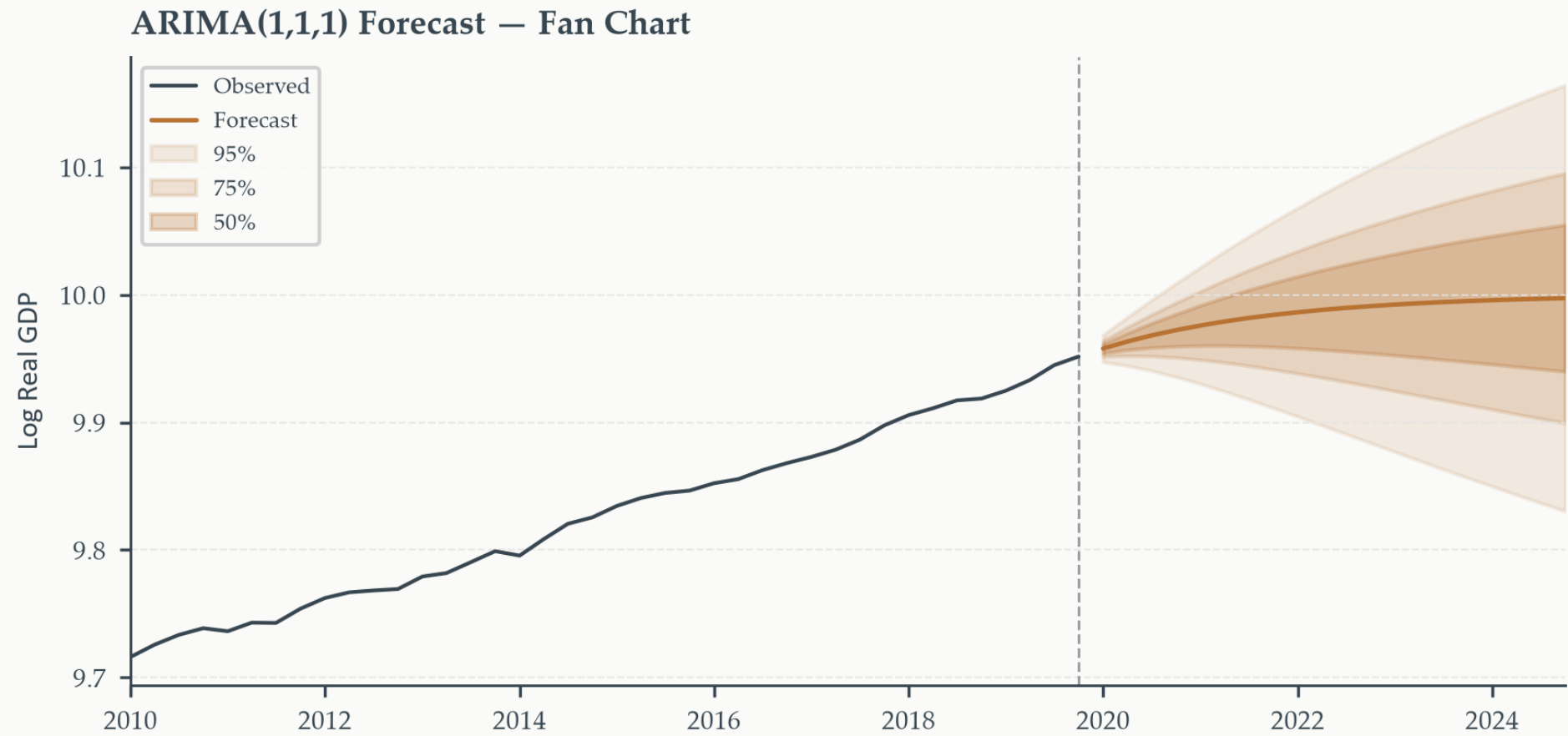
ARIMA Forecasting

- The ARIMA model forecasts changes, not levels
- You need to reconstruct $\hat{y}_{T+h|T}$ from estimated values of $\Delta\hat{y}_t$
- $$\hat{y}_{T+h|T} = y_T + \sum_{j=1}^h \Delta\hat{y}_{T+j|T}$$

ARIMA Forecasting



ARIMA Forecasting



SARIMA Models

Section 5

SARIMA Models

- SARIMA models include a seasonal component to the regression
 - Two types of seasonality
 - Deterministic
 - Stochastic
 - Let s be the number of seasons per year (quarterly: $s = 4$, monthly: $s = 12$)
 - $\Delta_s = 1 - L^s$
 - $\Delta_s y_t = y_t - y_{t-s}$

SARIMA Models

- The SARIMA model (assuming a quarterly series):

$$y_t = \bar{y} + \underbrace{(\phi_1 y_{t-1} + \dots + \phi_p y_{t-p})}_{\text{non-seasonal AR}(p)} + \underbrace{(\Phi_1 y_{t-4} + \dots + \Phi_P y_{t-4P})}_{\text{seasonal AR}(P)} + \underbrace{(\varepsilon_t + \theta_1 \varepsilon_{t-1} + \dots + \theta_q \varepsilon_{t-q})}_{\text{non-seasonal MA}(q)} + \underbrace{(\Theta_1 \varepsilon_{t-4} + \dots + \Theta_Q \varepsilon_{t-4Q})}_{\text{seasonal MA}(Q)}$$

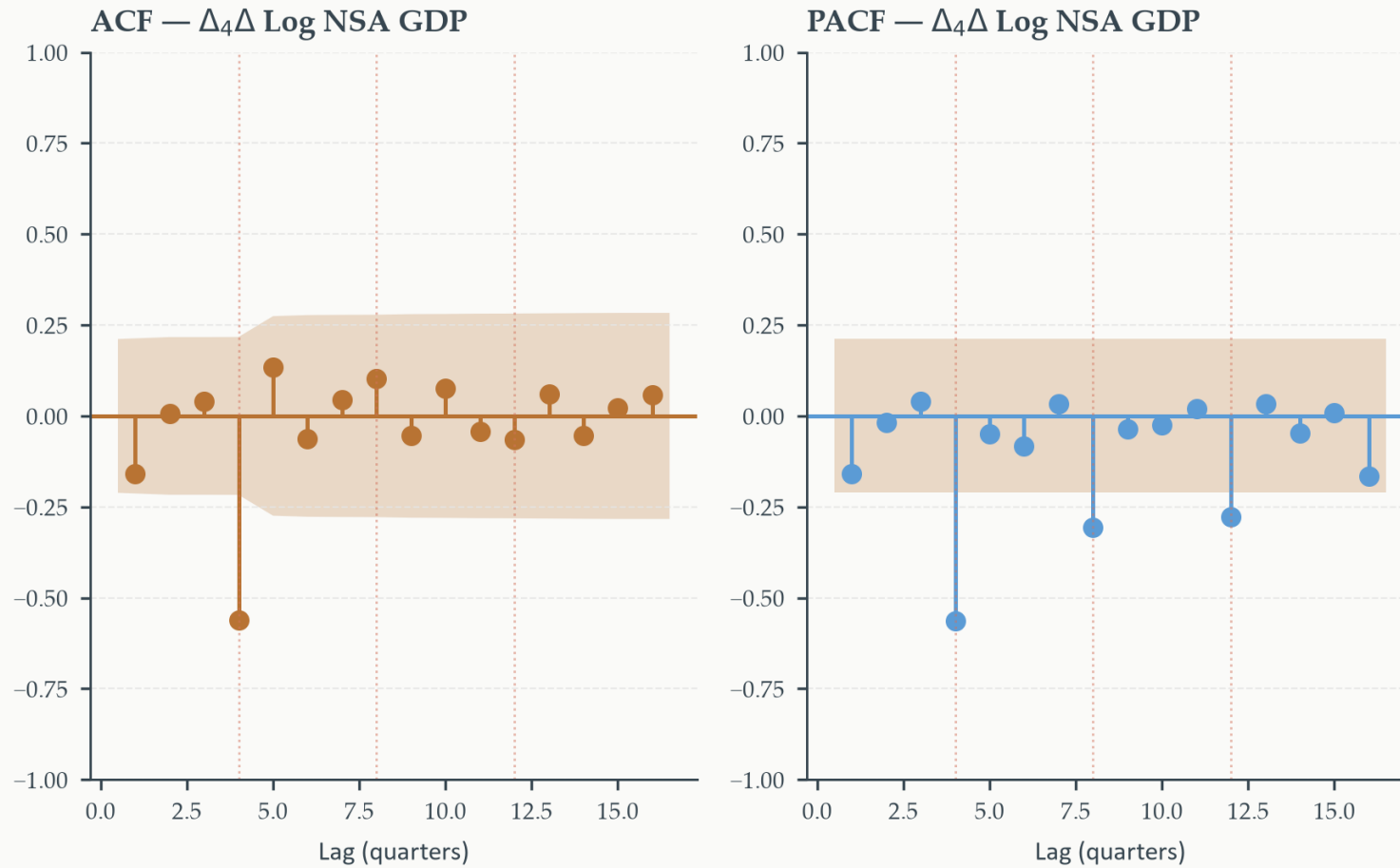
SARIMA Models

- How does seasonality look in the ACF/PACF plots?

ACF	PACF	Suggested structure
Tails off	Cuts off at lag p	ARIMA($p,d,0$)
Cuts off at lag q	Tails off	ARIMA($0,d,q$)
Tails off	Cuts off at lags P_s	SARIMA($p,d,0$)($P,d,0$)
Cuts off at lags Q_s	Tails off	SARIMA($0,d,q$)($0,d,Q$)

SARIMA Models

NSA GDP: ACF and PACF of $\Delta_4\Delta$ Log Level



Model	AIC	BIC	HQ
SARIMA(0,1,1)(0,1,1)_4	-478.4	-468.6	-474.5
SARIMA(1,1,0)(0,1,1)_4	-478.1	-468.3	-474.1
SARIMA(1,1,1)(1,1,1)_4	-479.1	-464.4	-473.2
SARIMA(1,1,0)(1,1,0)_4	-455.4	-445.6	-451.4

BIC-preferred: SARIMA(0,1,1)(0,1,1)_4

SARIMAX Results

```

=====
Dep. Variable:          Log NSA    No. Observations:           9
Model:                SARIMAX(0, 1, 1)x(0, 1, 1, 4)    Log Likelihood          243.20
Date:                  Mon, 14 Sep 2026    AIC                  -478.41
Time:                  14:47:05    BIC                  -468.60
Sample:                01-01-2002    HQIC                 -474.46
                        - 07-01-2024

```

Covariance Type: opg

```

=====
              coef      std err          z      P>|z|      [0.025      0.975]
-----
intercept    2.142e-05      0.000      0.045      0.964     -0.001      0.001
ma.L1        -0.1640      0.068     -2.416      0.016     -0.297     -0.031
ma.S.L4      -0.8249      0.104     -7.968      0.000     -1.028     -0.622
sigma2        0.0002    1.76e-05     11.046      0.000      0.000      0.000

```

```

=====
Ljung-Box (L1) (Q):           0.02    Jarque-Bera (JB):          1351.71
Prob(Q):                     0.90    Prob(JB):                0.00
Heteroskedasticity (H):       4.54    Skew:                    -2.60
Prob(H) (two-sided):          0.00    Kurtosis:                21.71
=====

```

The ETS-ARIMA Duality

Section 5

The ETS-ARIMA Duality

- The moving average methods assume a **fixed** weight or implicitly run an ARIMA model
- The ARIMA models **optimize** the weights
- The ETS models are particular cases of the ARIMA family

The ETS-ARIMA Duality

- Assume a one-step forward ARIMA(0,1,1) model
 - $\hat{y}_{t+1|t} = y_t + \theta_1 \hat{\varepsilon}_t, \quad \hat{\varepsilon}_t = y_t - \hat{y}_{t|t-1}$
 - $\hat{y}_{t+1|t} = y_t + \theta_1 (y_t - \hat{y}_{t|t-1})$
 - $\hat{y}_{t+1|t} = (1 + \theta_1)y_t - \theta_1 \hat{y}_{t|t-1}, \quad \alpha = 1 + \theta_1$
 - $\hat{y}_{t+1|t} = \alpha y_t + (1 - \alpha) \hat{y}_{t|t-1}$
 - This is the SES recursion from Chapter 1

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